

UPLOAD SLOT 12 OF 12 · ONE-MINUTE CONCEPT ANIMATION — STORYBOARD

Concept Animation — Storyboard

Sixty seconds, drawn from the project's own analysis

The film is not an illustration of the design; it is rendered from the design. Its geometry comes from the same arc definition as the masterplan and its sun from the same 8,760-hour solar model, so it cannot drift away from the drawings the way an animation produced separately would.

Al Safa 2 Park · **Falaj Al Safa** · 15,000 m² · Al Safa 2, Dubai · Mohamed Wasim, Individual Applicant · Issued 05 August 2026 · every quantity regenerated by `python run_analysis.py`

1. Structure — five scenes, sixty seconds

Time	Scene	What it shows
0–7 s	The site	Al Safa 2 as it is today: flat, exposed, and empty at midday
7–18 s	The heat	The reason it is empty — the modelled heat index across a summer day, and the hours that are lost to it
18–38 s	The crescent	The arc builds and plants itself while a full computed day passes over it. 131 trees, the falaj, the radial alleys and the rooms appear in the order they are reached
38–48 s	Beneath it	Eye level on the walk: the dappled soffit, the louvre cutting the low sun, the channel at the edge
48–60 s	The proof	The measured result — comfortable hours rising from 44.5% to 64.6% — as the park fills at dusk

Scene boundaries are asserted by `tests/test_film.js`, which renders every frame and fails on any gap, overlap or invalid value.

2. Why it can be trusted against the drawings

- The arc is the same 141 m radius as the masterplan, regenerated by `tools/sync_film.py` from `src/plan.py`.
- The sun positions are the same NREL-computed values used by the shade model — the shadows move correctly for the site's latitude and the time of year being shown.
- The tree count, the 0.9 m channel and the room layout are read from the same geometry as every drawing in this submission.
- Every frame is rendered by an automated test that fails on any NaN or undefined value, so a geometry change cannot break the film quietly.

3. Three cuts, one film

The submitted video is the hero cut. Two others exist because they answer different questions, and all three run sixty seconds against the same narration and the same figures.

Cut	What it shows	Built by
<code>concept_film_hero.html</code> the submitted film	Drawn entirely from the project's data — no photographs. Seven plan forms swept against the 8,760-hour solar model and scored, a real August day passing over the adopted arc, and a peak heat index falling from 56.8 °C to 48.7 °C.	<code>tools/build_concept_film_hero.py</code>
<code>concept_film_presentation.html</code>	The six photoreal renders, moving between each visualisation and the analysis behind it.	<code>tools/build_concept_film_v2.py</code>
<code>concept_film.html</code>	The park drawn in 3D in real time from <code>src/plan.py</code> with shadows computed live — evidence that the film and the drawings share one geometry.	hand-authored; geometry by <code>tools/sync_film.py</code>

All three record their own MP4. Every frame of the third is asserted by `tests/test_film.js`.

The hero cut is submitted because the plan-form sweep is the part of this proposal that cannot be photographed. The straight bar shades more ground on average and the film shows it doing so; the arc is adopted on hours in the year when the route offers nowhere at all to stand, 330 down to 52. That is the clearest available answer to how AI changed the design rather than decorated it.

3. The narration

The film carries a spoken commentary in four fifteen-second segments, cued to 0, 15, 30 and 45 seconds. The voice is synthesised; the words are not. Every figure it speaks — 56.8 °C in the open, a 141 m radius, 44.5% rising to 64.6% — is the same figure the analysis produces and the same one printed elsewhere in this submission.

The audio is embedded in the page rather than kept in files beside it. Opened as a local file, a browser refuses to load a sibling audio file at all, and cannot route one into a recording; inlined, it does both. That also keeps the film a single standalone file, which is the same reason its geometry is inlined rather than fetched.

4. How to produce the video file

The film records itself. Every frame is a pure function of one number — the time in seconds — so the page can hand the canvas and the narration straight to the browser's recorder and write the video out. Nothing is screen-captured, so the file carries no browser chrome, no desktop behind it, and no dependence on what the window was scaled to.

- Open `concept_film.html` in Chrome or Edge.
- Press **Record the film to a video file** and leave the tab in front for sixty seconds.
- **Quality** beside it switches the recording between 1920×1080 and 3840×2160. At 4K the film's backing canvas is doubled and the drawing context scaled, so every line and glyph is resolved at 4K rather than upscaled from 1080p — the bitrate rises with it.
- The file downloads on its own as `Falaj_AI_Safa_Concept_Film_60s_4K.mp4` — sixty seconds with narration, about 43 MB at 4K or 28 MB at 1080p.
- **Narration on / off** beside it records a silent version instead, if a voice-over is not wanted.

Browsers that cannot write MP4 fall back to WebM, and a WebM written this way carries no duration in its header — it plays, but some players show no timeline. Convert it with `ffmpeg -i film.webm -c:v libx264 -pix_fmt yuv420p film.mp4`, or record again in Chrome or Edge, which write MP4 directly.

The film shows the **analysed** scheme exactly: the same arc, the same trees, the same solar model. It is therefore consistent with every number quoted elsewhere in this submission.